脳神経外科講義　水頭症（成人）脳脊髄液動態の転換

山田 茂樹
名古屋市立大学 脳神経外科学講座

Third Circulation Theory
（第三の循環仮説）

CSFは、脈絡叢で1日に約500 mL 産生される.

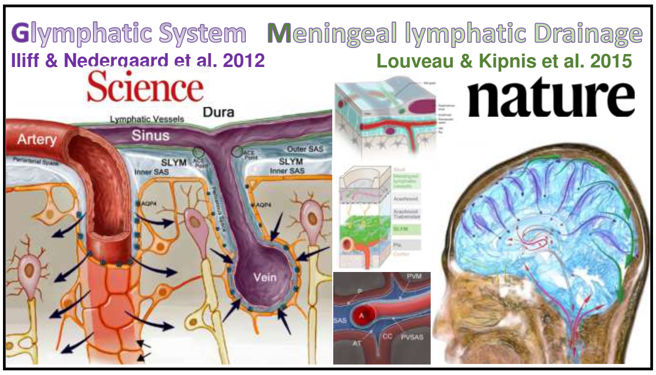
頭蓋内には約150mL 存在する.

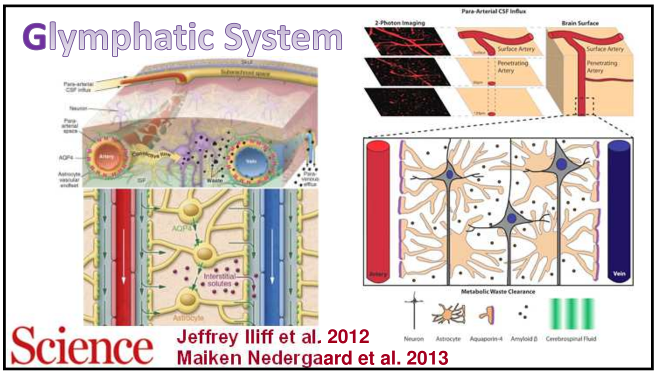
側脳室 ➡ 第三脳室 ➡ 第四脳室へと流れ、

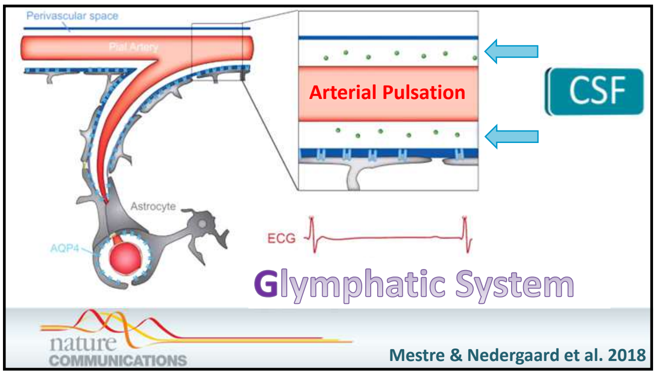
Luschka孔・Magendie孔を通り、

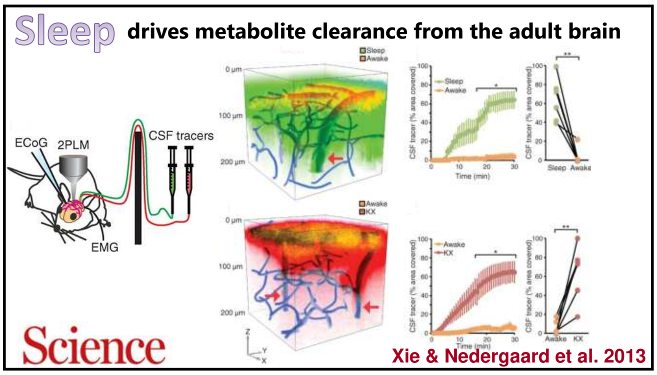
くも膜下腔に出て、SSSへ向かって上行する.

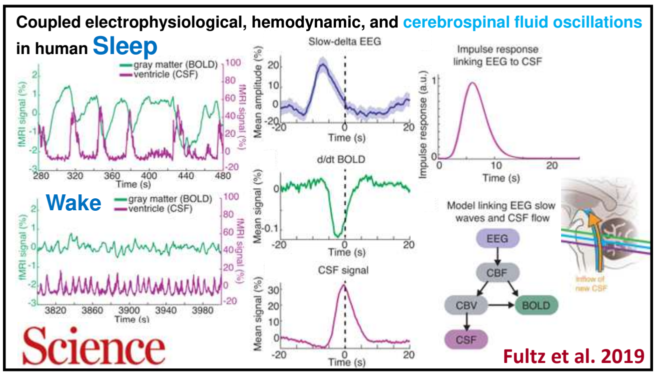
最終的に、くも膜顆粒から吸収される.

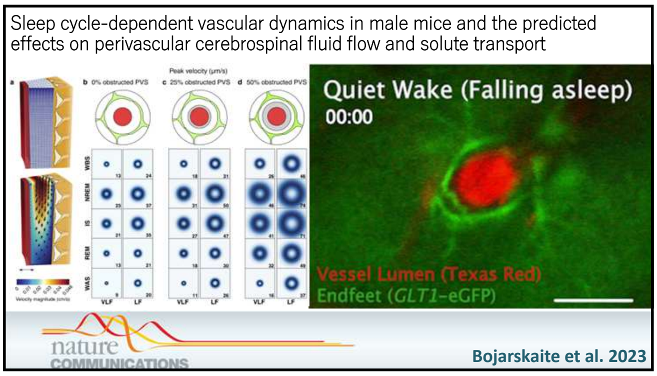


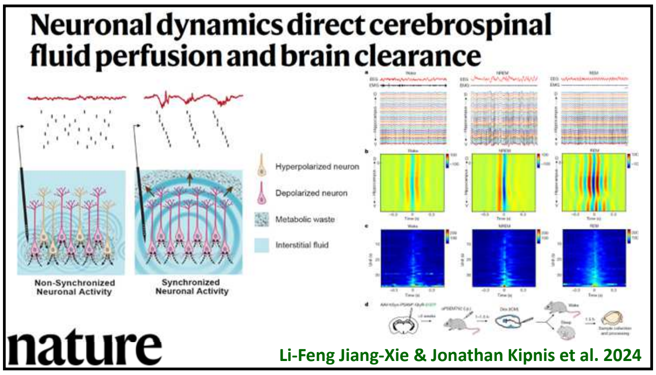


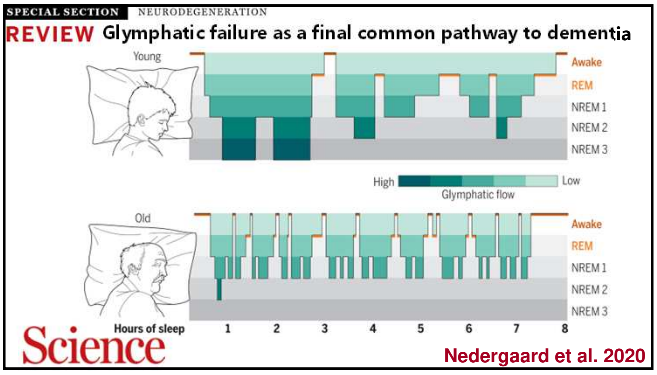


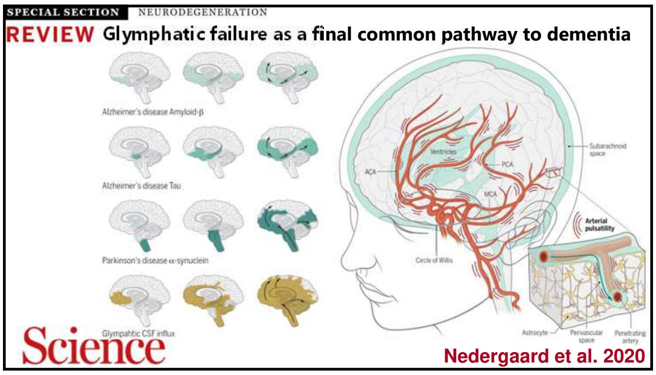


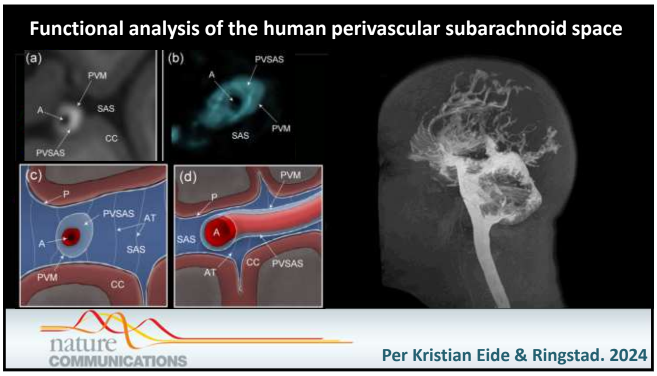

Coupled electrophysiological, hemodynamic, and cerebrospinal fluid oscillations in human Sleep
Wake
gray matter (BOLD)
ventricle (CSF)
Slow-delta EEG
d/dt BOLD
CSF signal
Impulse response linking EEG to CSF
Model linking EEG slow waves and CSF flow
EEG
CBF
CBV
BOLD
CSF
Science
Fultz et al. 2019

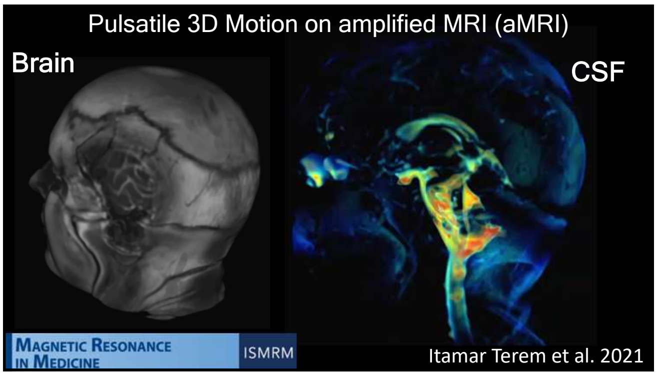

Sleep cycle-dependent vascular dynamics in male mice and the predicted effects on perivascular cerebrospinal fluid flow and solute transport
Quiet Wake (Falling asleep)
00:00
Vessel Lumen (Texas Red)
Endfeet (GLT1-eGFP)
nature COMMUNICATIONS
Bojarskaite et al. 2023

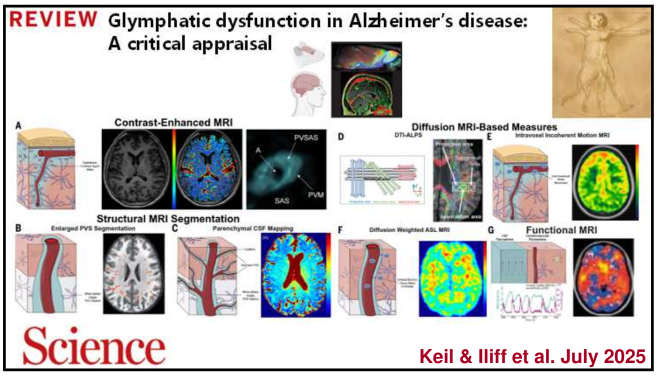

Neuronal dynamics direct cerebrospinal fluid perfusion and brain clearance
Non-Synchronized Neuronal Activity
Synchronized Neuronal Activity
Hyperpolarized neuron
Depolarized neuron
Metabolic waste
Interstitial fluid
nature
Li-Feng Jiang-Xie & Jonathan Kipnis et al. 2024


SPECIAL SECTION NEURODEGENERATION
REVIEW Glymphatic failure as a final common pathway to dementia
Young
Old
Awake
REM
NREM 1
NREM 2
NREM 3
High
Low
Glymphatic flow
Hours of sleep
1 2 3 4 5 6 7 8
Science
Nedergaard et al. 2020

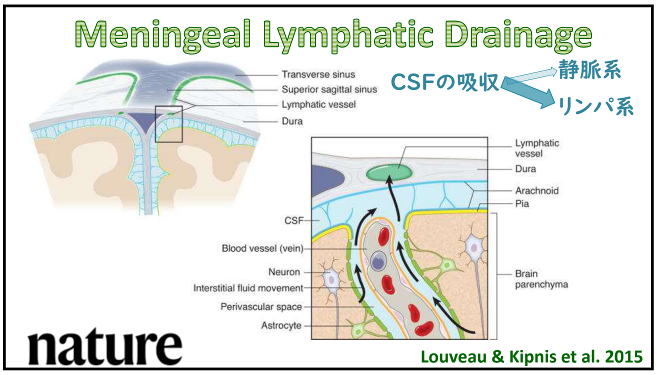

SPECIAL SECTION NEURODEGENERATION
REVIEW Glymphatic failure as a final common pathway to dementia
Alzheimer's disease Amyloid-β
Alzheimer's disease Tau
Parkinson's disease α-synuclein
Glymphatic CSF influx
Subarachnoid space
Arterial pulsatility
Circle of Willis
Astrocyte
Perivascular space
Penetrating artery
Science
Nedergaard et al. 2020

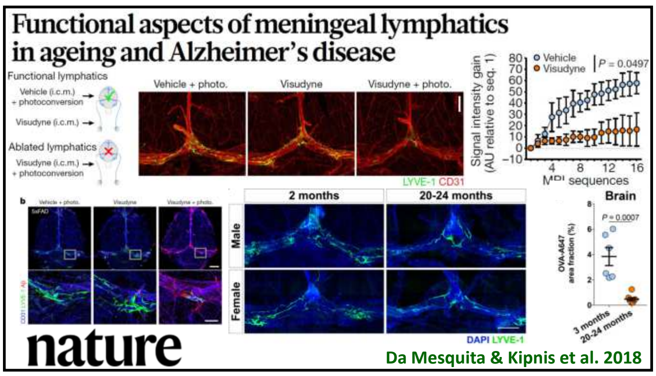

Functional analysis of the human perivascular subarachnoid space
(a)
(b)
(c)
(d)
PVSAS
PVM
SAS
CC
AT
nature COMMUNICATIONS
Per Kristian Eide & Ringstad. 2024

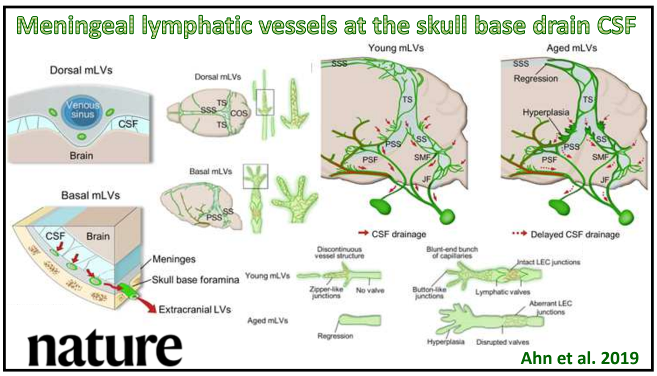

Pulsatile 3D Motion on amplified MRI (aMRI)
Brain
CSF
Magnetic Resonance in Medicine
ISMRM
Itamar Terem et al. 2021

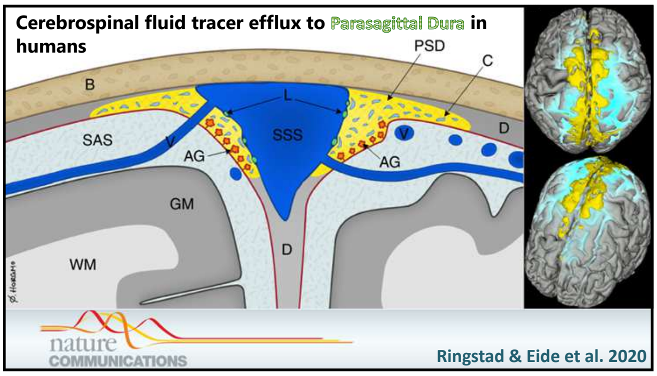

REVIEW
Glymphatic dysfunction in Alzheimer's disease: A critical appraisal
Contrast-Enhanced MRI
Diffusion MRI-Based Measures
Structural MRI Segmentation
Functional MRI
Science
Keil & Iliff et al. July 2025

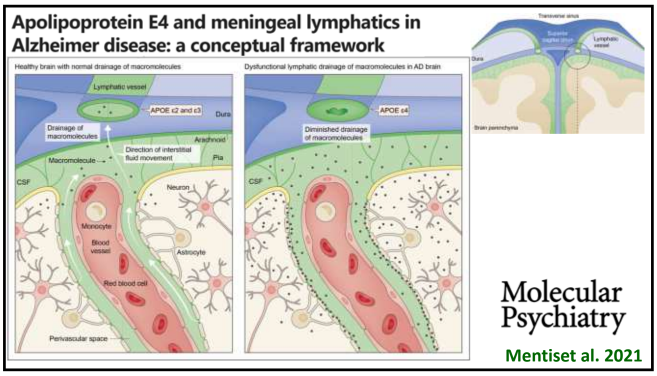

Does sleep really clean the brain?
Maybe not, new paper argues
A mouse study challenges popular "glymphatic" theory, but its methodology is drawing criticism
Science
2024

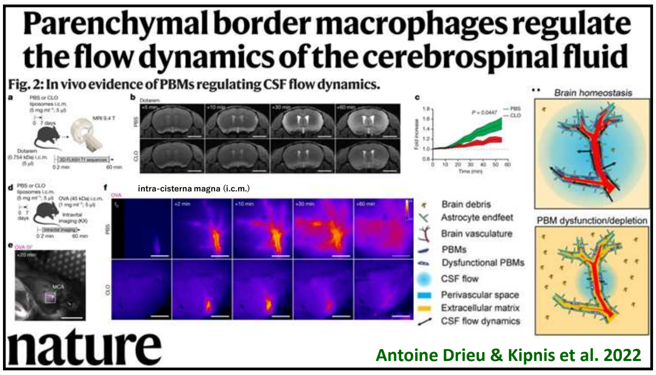

Meningeal Lymphatic Drainage
Transverse sinus
Superior sagittal sinus
Lymphatic vessel
Dura
CSFの吸収
静脈系
リンパ系
Lymphatic vessel
Dura
Arachnoid
Pia
CSF
Blood vessel (vein)
Neuron
Interstitial fluid movement
Perivascular space
Astrocyte
Brain parenchyma
nature
Louveau & Kipnis et al. 2015

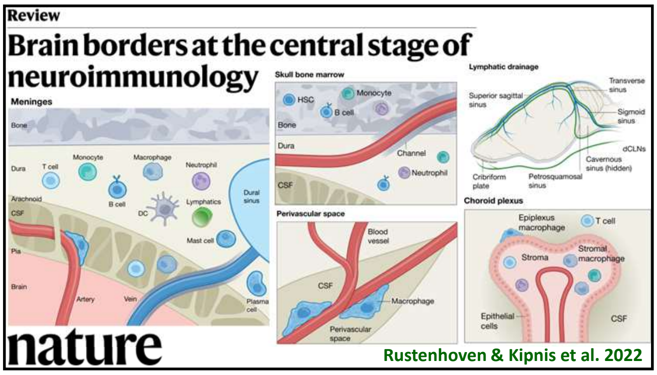

Functional aspects of meningeal lymphatics in ageing and Alzheimer's disease
Functional lymphatics
Ablated lymphatics
Vehicle + photo.
Visudyne
Visudyne + photo.
LYVE-1 CD31
MRI sequences
2 months
20-24 months
Brain
Male
Female
DAPI LYVE-1
nature
Da Mesquita & Kipnis et al. 2018

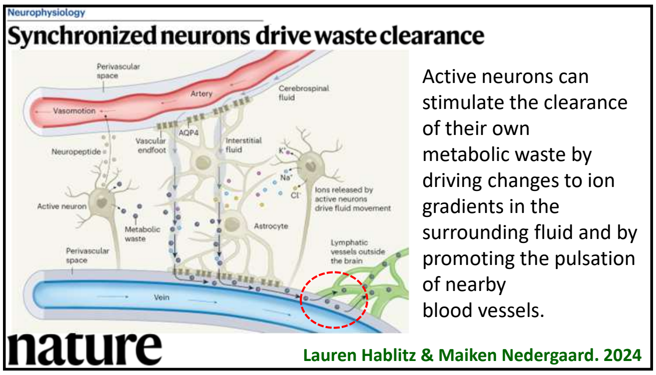

Meningeal lymphatic vessels at the skull base drain CSF
Dorsal mLVs
Basal mLVs
Young mLVs
Aged mLVs
Venous sinus
CSF
Brain
Meninges
Skull base foramina
Extracranial LVs
CSF drainage
Delayed CSF drainage
Regression
Hyperplasia
nature
Ahn et al. 2019

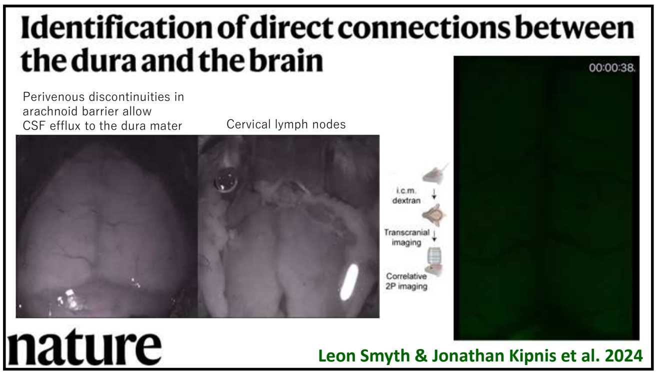

Cerebrospinal fluid tracer efflux to Parasagittal Dura in humans
PSD
C
B
L
SSS
SAS
AG
AG
V
D
GM
WM
D
nature COMMUNICATIONS
Ringstad & Eide et al. 2020


Apolipoprotein E4 and meningeal lymphatics in Alzheimer disease: a conceptual framework
Healthy brain with normal drainage of macromolecules
Dysfunctional lymphatic drainage of macromolecules in AD brain
Lymphatic vessel
APOE ε2 and ε3
APOE ε4
Dura
Arachnoid
Pia
CSF
Neuron
Astrocyte
Monocyte
Blood vessel
Red blood cell
Perivascular space
Molecular Psychiatry
Mentiset al. 2021


Parenchymal border macrophages regulate the flow dynamics of the cerebrospinal fluid
Fig. 2: In vivo evidence of PBMs regulating CSF flow dynamics.
intra-cisterna magna (i.c.m.)
Brain homeostasis
PBM dysfunction/depletion
Brain debris
Astrocyte endfeet
Brain vasculature
PBMs
Dysfunctional PBMs
CSF flow
Perivascular space
Extracellular matrix
CSF flow dynamics
nature
Antoine Drieu & Kipnis et al. 2022

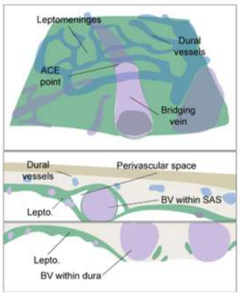

Review
Brain borders at the central stage of neuroimmunology
Meninges
Skull bone marrow
Lymphatic drainage
Perivascular space
Choroid plexus
HSC
Monocyte
B cell
T cell
Macrophage
Neutrophil
Lymphatics
Mast cell
Dural sinus
Plasma cell
Artery
Vein
Blood vessel
Macrophage
Superior sagittal sinus
Transverse sinus
Sigmoid sinus
dCLNs
Cribriform plate
Petrosquamosal sinus
Cavernous sinus (hidden)
Epiplexus macrophage
Stroma
Stromal macrophage
Epithelial cells
CSF
nature
Rustenhoven & Kipnis et al. 2022


Neurophysiology
Synchronized neurons drive waste clearance
Perivascular space
Artery
Cerebrospinal fluid
Vasomotion
AQP4
Vascular endfoot
Interstitial fluid
Neuropeptide
Active neuron
Metabolic waste
Astrocyte
Ions released by active neurons drive fluid movement
Perivascular space
Lymphatic vessels outside the brain
Vein
Active neurons can stimulate the clearance of their own metabolic waste by driving changes to ion gradients in the surrounding fluid and by promoting the pulsation of nearby blood vessels.
nature
Lauren Hablitz & Maiken Nedergaard. 2024


Identification of direct connections between the dura and the brain
Perivenous discontinuities in arachnoid barrier allow CSF efflux to the dura mater
Cervical lymph nodes
i.c.m. dextran
Transcranial imaging
Correlative 2P imaging
00:00:38
nature
Leon Smyth & Jonathan Kipnis et al. 2024

Identification of direct connections between the dura and the brain

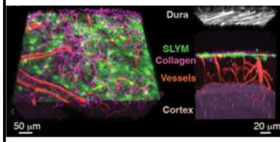
Movie 4: Volume rendering of a bridging vein exiting the subarachnoid space into the dura in a Cdh5-CreERT2::tdTom mouse.

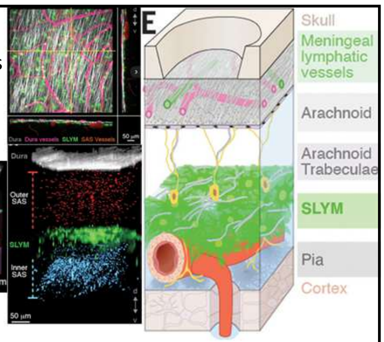


nature

Leon Smyth & Jonathan Kipnis et al. 2024

Subarachnoid Lymphatic-like Membrane (SLYM) represents a barrier that subdivides the subarachnoid space into two functional compartments.

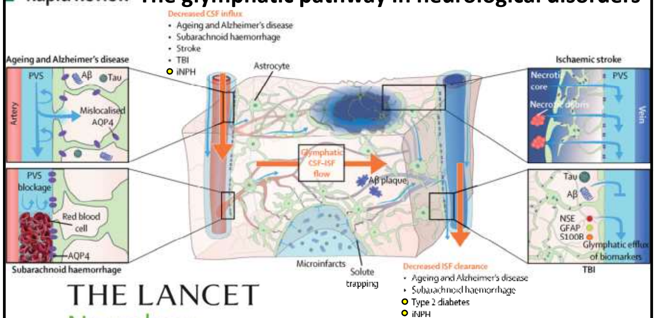




Science

Mollgard & Nedergaard et al. 2023

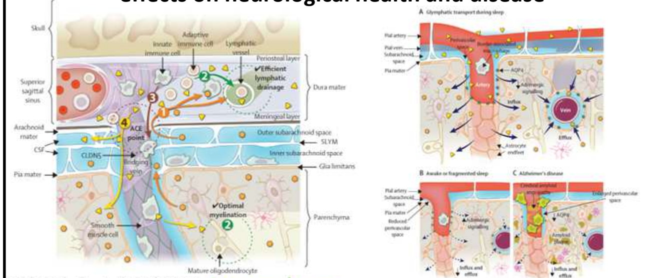
Rapid Review The glymphatic pathway in neurological disorders



THE LANCET Neurology

Rasmussen & Nedergaard et al. 2018

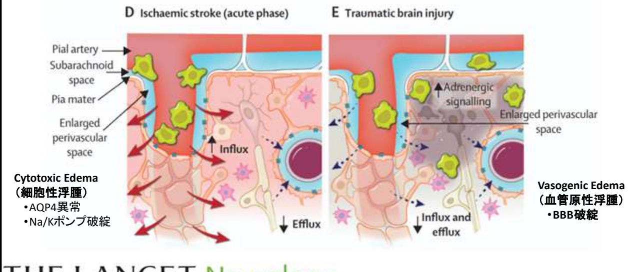
Review CSF transport at the brain–meningeal border: effects on neurological health and disease



THE LANCET Neurology

Pinho-Correia & Nedergaard et al. 2025

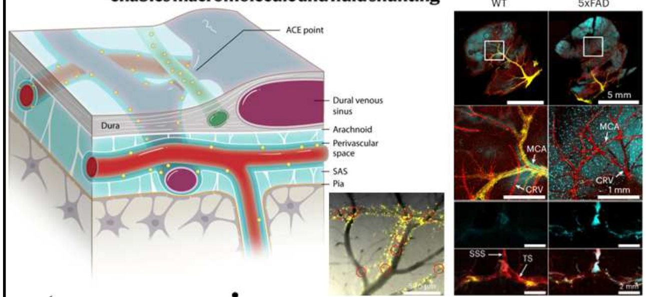
Review CSF transport at the brain–meningeal border: effects on neurological health and disease



THE LANCET Neurology

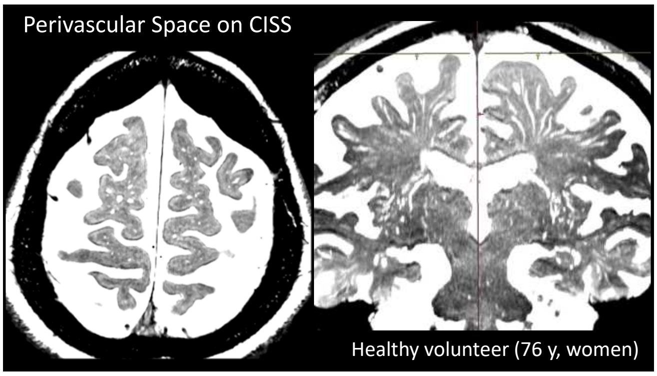
Pinho-Correia & Nedergaard et al. 2025

A route for cerebrospinal fluid flow through leptomeningeal arterial–venous overlaps enables macromolecule and fluid shunting

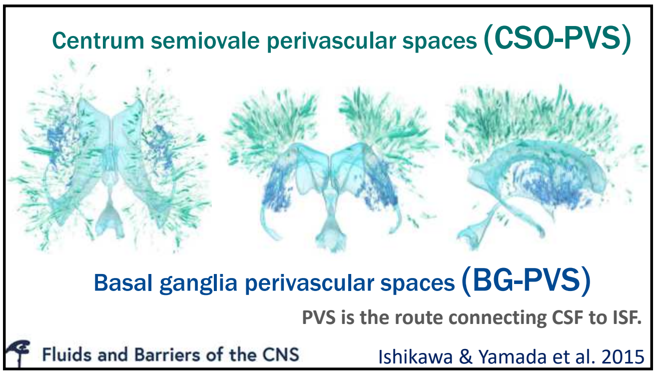


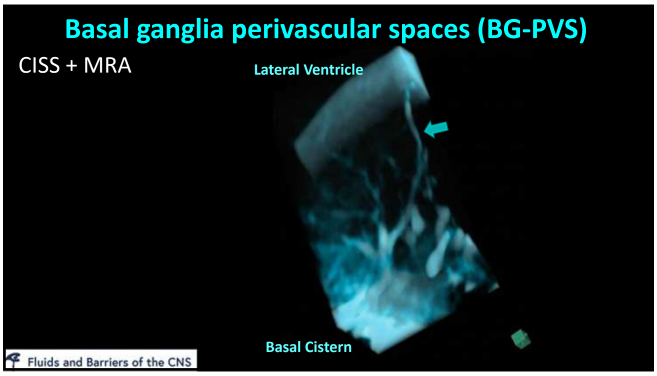
nature neuroscience

Benjamin Plog & Kipnis et al. April 2025

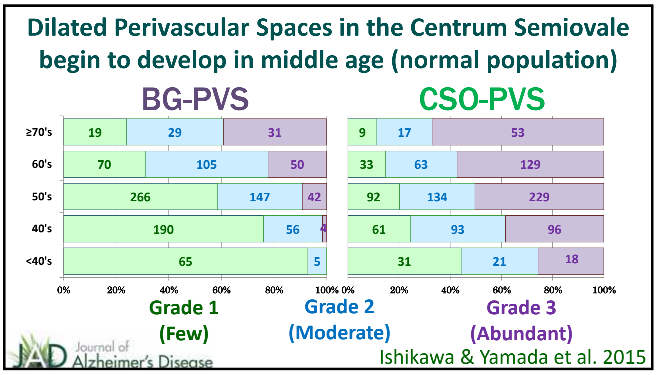

Perivascular Space on CISS
Healthy volunteer (76 y, women)

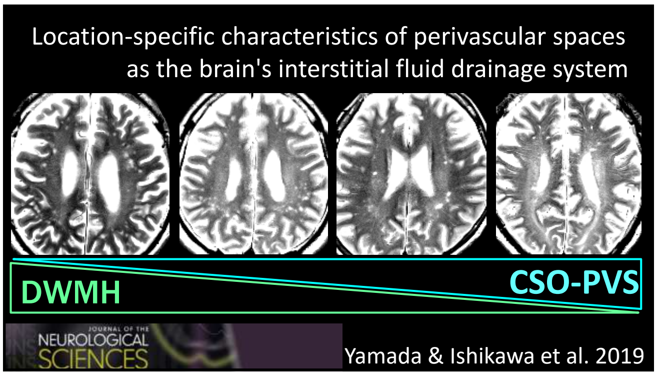

The Row Boat
1887
The letter to his friend, Gustave Geffroy
22 June 1890
"Water, with weeds waving at the bottom. It is a wonderful sight, but it drives one to crazy to try to paint it.
But that is the kind of thing I am always a tackling."
Claude Monet
(1840 – 1926)

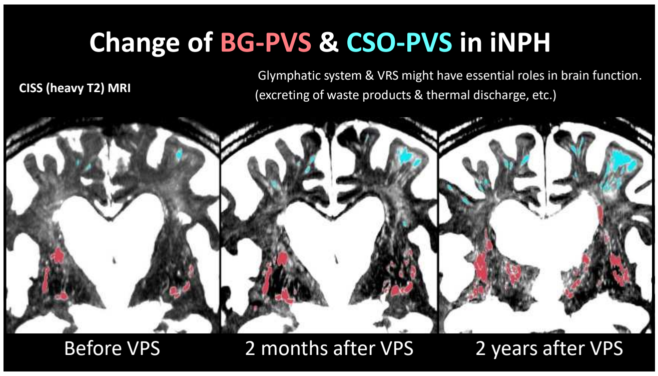

Centrum semiovale perivascular spaces (CSO-PVS)
Basal ganglia perivascular spaces (BG-PVS)
PVS is the route connecting CSF to ISF.
Fluids and Barriers of the CNS
Ishikawa & Yamada et al. 2015

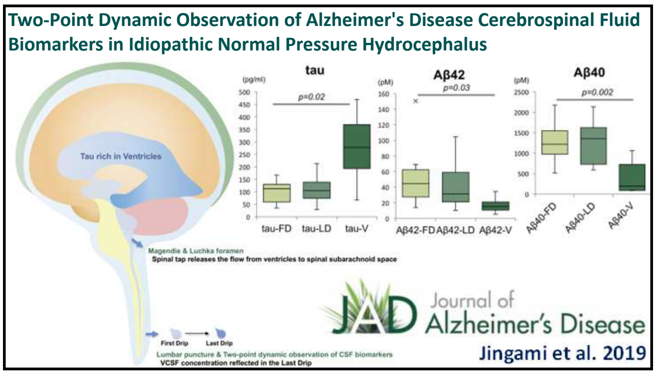

Basal ganglia perivascular spaces (BG-PVS)
CISS + MRA
Lateral Ventricle
Basal Cistern
Fluids and Barriers of the CNS

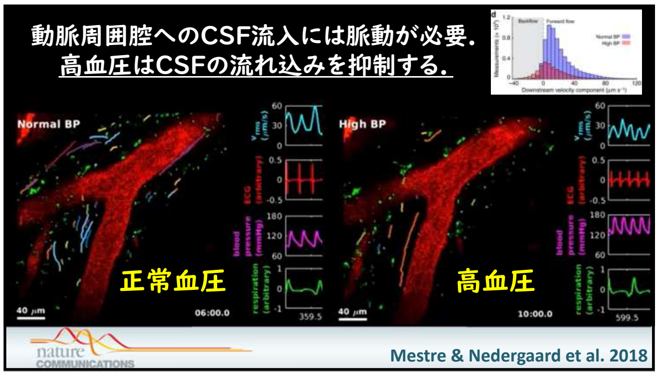

Centrum Semiovale Perivascular Spaces (CSO-PVS)
Fluids and Barriers of the CNS

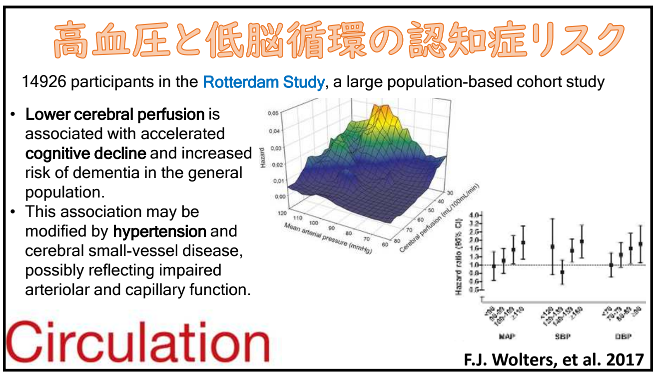

Dilated Perivascular Spaces in the Centrum Semiovale begin to develop in middle age (normal population)
BG-PVS
CSO-PVS
≥70's 19 29 31
60's 70 105 50
50's 266 147 42
40's 190 56 4
<40's 65 5
9 17 53
33 63 129
92 134 229
61 93 96
31 21 18
0% 20% 40% 60% 80% 100%
Grade 1 (Few)
Grade 2 (Moderate)
Grade 3 (Abundant)
Journal of Alzheimer's Disease
Ishikawa & Yamada et al. 2015











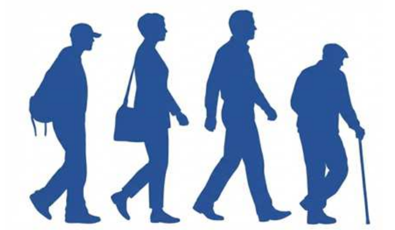
良い睡眠

深く・長く ＋ 正常血圧

睡眠障害＋（夜間）高血圧

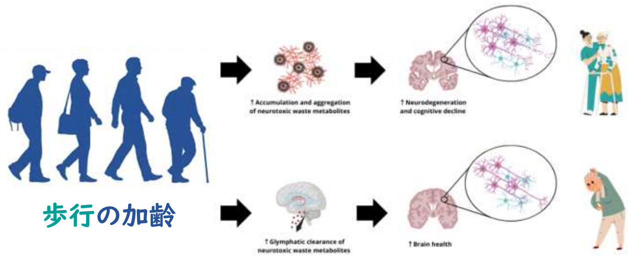
認知症リスク

健康のためのウォーキング



歩行の加齢

歩行と良い睡眠で脳をきれいに洗浄！



従来の推奨：毎週150～300分（1日20分以上）の適度な身体活動

最近の研究：毎週15分（1日平均2.2分）の激しい身体活動

運動スナック

(上り坂やランニングなど、話すのが難しい運動)

"Move more, reduce sedentary time, sleep well"

もっと動いて、座っている時間を減らし、良く眠ろう！

"Doing some is better than doing none"

何もしないより、何かをする方がましだ！

nature

Fitbit Inspire 3

心拍数・心拍変動・呼吸回数

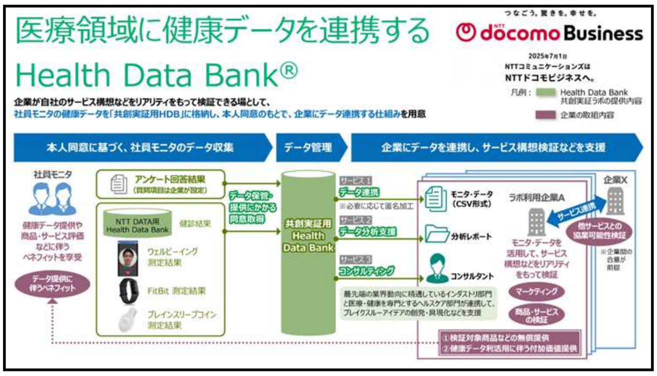
歩数・歩行距離・消費カロリー

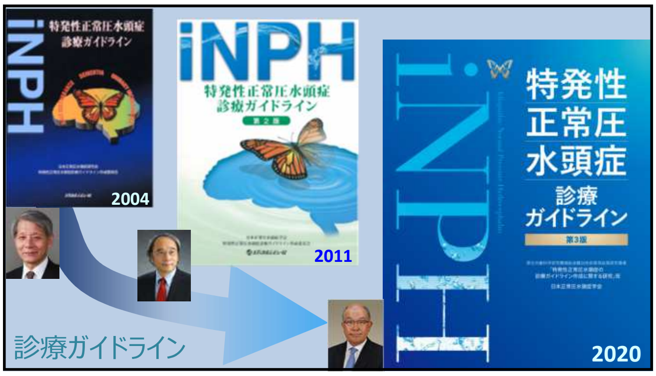
Fitbit Inspire 3 睡眠計測

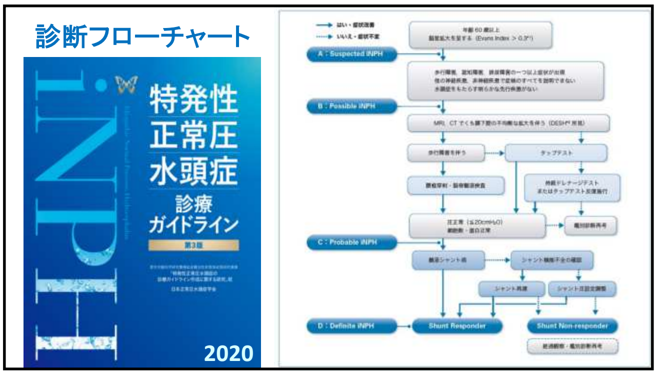
Apple Watch

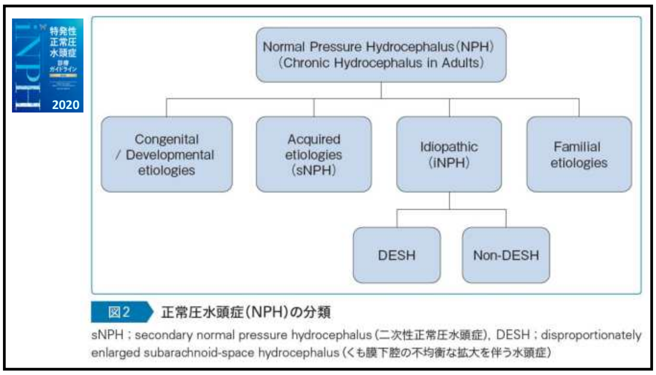
スマホ内に保存されるヘルスケアデータ活用


わたしたちは、さりげない健康を提供します。
さりげない見守り、さりげない健康づくりが、なにげない日常を豊かな毎日にする、そんな社会を築いていきます。
楽しみながら健康づくり
・歩数や体重、血圧などのバイタルデータの健康管理
・マイナポータル連携により健診結果を取得可能
・ウォーキングコースや写真投稿などの楽しいコンテンツ
・電子マネーなどのインセンティブ付与等に対応
さりげない健康づくり
ウォーキングを楽しく続けて「健康で長生き」をめざす！
ささえあうつなぐ
導入自治体数
累計151自治体
累計利用者数
75万人以上
けんこうマイレージ
HealthDataBank
NTT DATA
つなごう。驚きを。幸せを。
docomo Business


医療領域に健康データを連携する
Health Data Bank®
docomo Business
2025年7月1日 NTTコミュニケーションズは NTTドコモビジネスへ。
企業が自社のサービス構想などをリアリティをもって検証できる場として、
社員モニタの健康データを「共創実証用HDB」に格納し、本人同意のもとで、企業にデータ連携する仕組みを用意
本人同意に基づく、社員モニタのデータ収集
データ管理
企業にデータを連携し、サービス構想検証などを支援
社員モニタ
アンケート回答結果
NTT DATA用 Health Data Bank
健診結果
ウェルビーイング測定結果
FitBit 測定結果
共創実証用 Health Data Bank
モニタ・データ（CSV形式）
分析レポート
コンサルタント
ラボ利用企業A
企業X


iNPH
特発性正常圧水頭症診療ガイドライン
2004
iNPH
特発性正常圧水頭症診療ガイドライン
第2版
2011
iNPH
特発性正常圧水頭症診療ガイドライン
第3版
2020
診療ガイドライン


診断フローチャート
iNPH
特発性正常圧水頭症診療ガイドライン
第3版
2020
A : Suspected iNPH
B : Possible iNPH
C : Probable iNPH
D : Definite iNPH
Shunt Responder
Shunt Non-responder

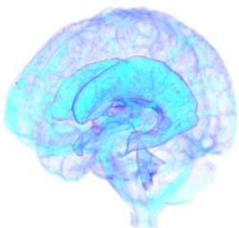

iNPH
特発性正常圧水頭症診療ガイドライン
2020
Normal Pressure Hydrocephalus (NPH)
(Chronic Hydrocephalus in Adults)
Congenital / Developmental etiologies
Acquired etiologies (sNPH)
Idiopathic (iNPH)
Familial etiologies
DESH
Non-DESH
図2 正常圧水頭症（NPH）の分類
sNPH：secondary normal pressure hydrocephalus（二次性正常圧水頭症）, DESH：disproportionately enlarged subarachnoid-space hydrocephalus（くも膜下腔の不均衡な拡大を伴う水頭症）


2025/
The New York Times
2025/5/24
Billy Joel Announces Brain Disorder and Cancels All Concerts
Joel said he had normal pressure hydrocephalus, which has led to "problems with hearing, vision and balance."
Billy Joel Has Normal Pressure Hydrocephalus. What Is It?
The singer canceled his upcoming concerts because of the brain disorder. Here's what to know about the symptoms, prognosis and treatment.

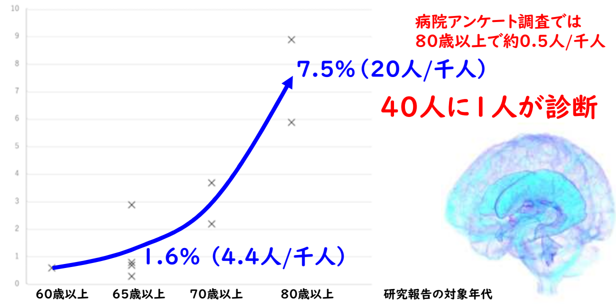
目標：Hakim病（iNPH）を適切に治療する

不確実な診断・治療となりやすい理由

1. 認知度が低い。
2. 手術しても治らないだろう。
3. 脳萎縮と見誤られる。



ハキム病（iNPH）患者の割合（%）



INVITED REVIEW ARTICLE

Neurologia i Neurochirurgia Polska
Polish Journal of Neurology and Neurosurgery
2023, DOI: 10.5603/pjnns.97343

Normal pressure hydrocephalus, or Hakim syndrome: review and update

Philip W. Tipton[1], Benjamin D. Elder[2], Petrice M. Cogswell[3], Neill Graff-Radford[1]
[1]Department of Neurology, Mayo Clinic, Jacksonville, Florida, United States
[2]Department of Neurosurgery, Mayo Clinic, Rochester, Minnesota, United States
[3]Department of Neuroradiology, Mayo Clinic, Rochester, Minnesota, United States

NPH and iNPH are misnomers.

誤称

国際水頭症学会 2024
プレミーティングセミナー
in 名古屋

Hydrocephalus Renaming Task Force Team

LITERATURE REVIEW

Classification of Chronic Hydrocephalus in Adults: A Systematic Review and Analysis

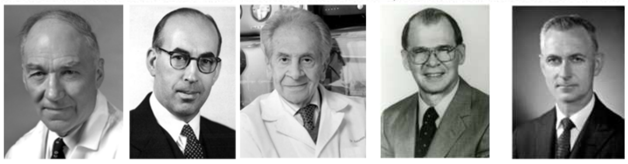
Mats Tullberg[1], Ahmed K. Toma[2], Shigeki Yamada[3], Katarina Laurell[4], Masakazu Miyajima[5], Laurence D. Watkins[2], Carsten Wikkelsø[1]

WORLD NEUROSURGERY 183: 113-122, MARCH 2024

The NEW ENGLAND JOURNAL of MEDICINE **1965**

Symptomatic Occult Hydrocephalus with Normal Cerebrospinal-Fluid Pressure — A Treatable Syndrome

R. D. Adams, M.D.†, C. M. Fisher, M.D.‡, S. Hakim, M.D.§, R. G. Ojemann, M.D.¶, and W. H. Sweet, M.D.‖



1911 – 2008	1913 – 2012	1922 – 2011	1931 – 2010	1910 – 2001
54歳	52歳	43歳	34歳	54歳

2010 CNS Neurosurgery

HISTORICAL LIBRARY

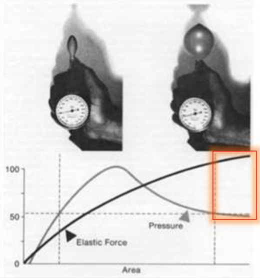
Salomón Hakim and the Discovery of Normal-Pressure Hydrocephalus

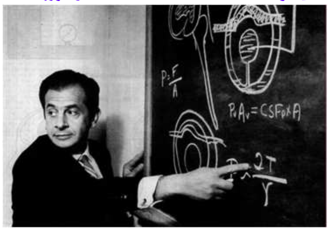
Neurological Surgery, The Neurological Institute, Columbia University, New York, USA
Wallenstein, Matthew B. MD; McKhann, Guy M. II MD

INTRODUCTION: Normal-pressure hydrocephalus (NPH) is a chronic neurological disorder characterized by enlarged ventricles and a triad of clinical symptoms affecting gait, cognition, and urinary continence. Salomón Hakim first identified the syndrome in 1957 at the Hospital San Juan de Dios in Bogotá, Colombia. Even after decades of international focus and thousands of publications on his disorder, Hakim's story remains largely untold.
METHODS: In this historical review, we explore the discovery of NPH through a series of personal interviews with Professor Hakim and his family, discussions with former colleagues, and review of the relevant medical literature.

なぜ脳室に圧力をかけないまま拡大させられるのか？
風船の**圧力**と**弾性力**の関係

静水圧　パスカルの原理

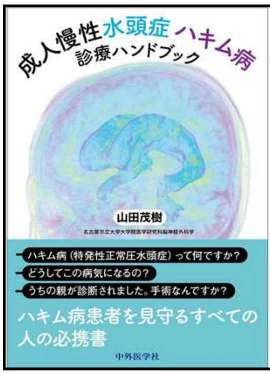




Dr. Salomón Hakim

1817年 ⟹ 55年 ⟹ 1872年
Dr. James Parkinson (England)
パーキンソン病
Prof. Jean-Martin Charcot (France)

1906年 ⟹ 5年 ⟹ 1910年
Dr. Alois Alzheimer (Germany)
アルツハイマー病
Prof. Emil Kraepelin (Germany)



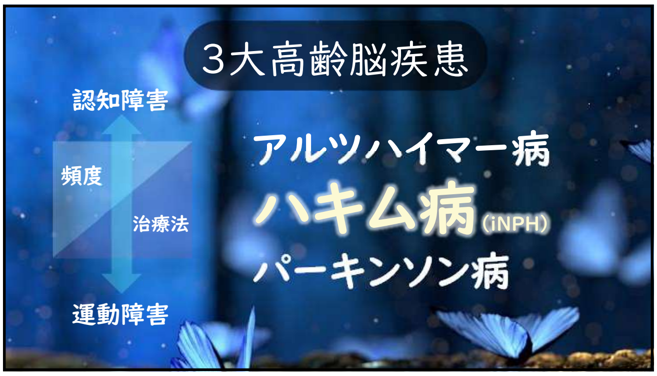
はじめに
CHAPTER 1 特発性正常圧水頭症（iNPH）の歴史
CHAPTER 2 ハキム病の命名
CHAPTER 3 ハキム病の頻度（疫学）
CHAPTER 4 ハキム病の発症機序
CHAPTER 5 ハキム病に特徴的な症状の総合評価
CHAPTER 6 ハキム病に特徴的な歩行障害
CHAPTER 7 ハキム病に特徴的な認知障害と排尿障害
CHAPTER 8 画像検査
CHAPTER 9 髄液排除試験（タップテスト）
CHAPTER 10 鑑別診断と併存疾患
CHAPTER 11 治療方法
CHAPTER 12 合併症
CHAPTER 13 髄液シャント術後のマネージメント
CHAPTER 14 ハキム病に関連する我が国の医療制度と医療資源
CHAPTER 15 特に印象に残っているハキム病患者さんたち
あとがき

2024/10/10 発売

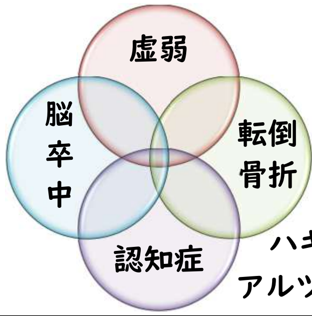
￥3,630 税込

日テレ「医療関係者でも知らない人が多い」手術で"改善が期待できる認知症"

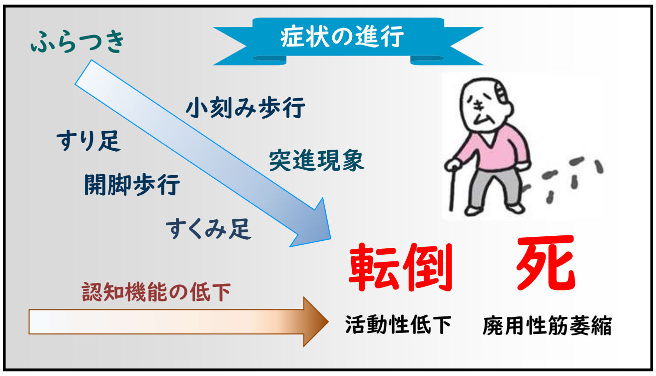
ハキム病とはー『every.特集』　2025/6/4



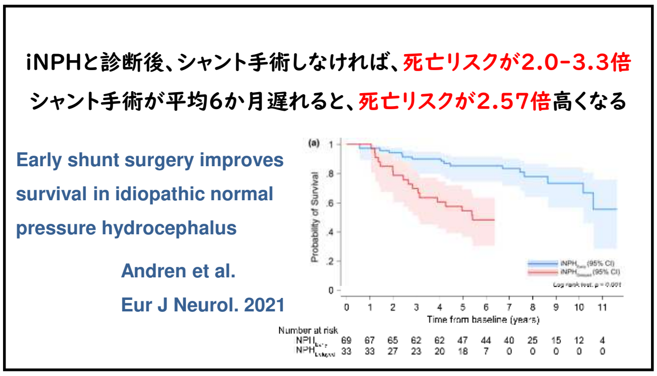
高齢者が要介護になる4大疾患



運動障害
パーキンソン病
ハキム病（iNPH）
アルツハイマー病

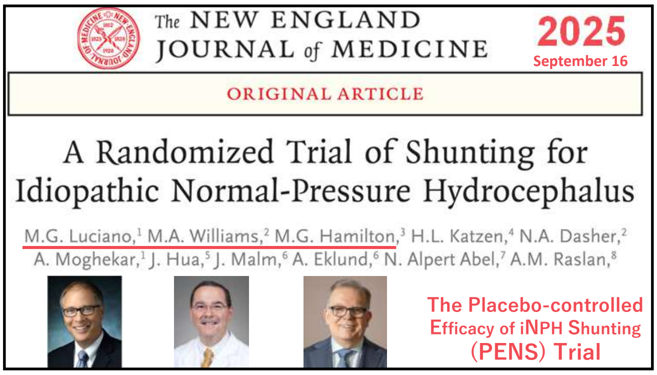


ふらつき
症状の進行
小刻み歩行
すり足
突進現象
開脚歩行
すくみ足
転倒
死
認知機能の低下
活動性低下
廃用性筋萎縮

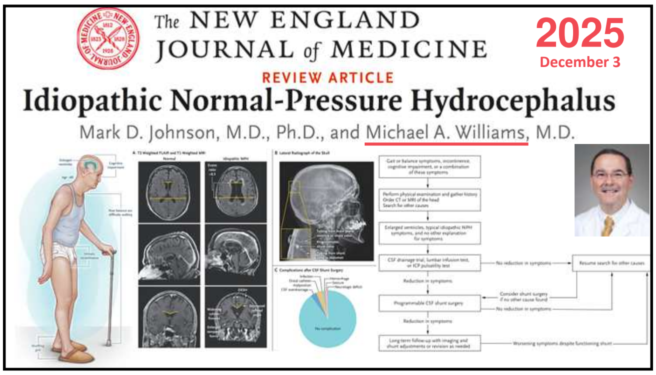


iNPHと診断後、シャント手術しなければ、死亡リスクが2.0-3.3倍
シャント手術が平均6か月遅れると、死亡リスクが2.57倍高くなる
Early shunt surgery improves survival in idiopathic normal pressure hydrocephalus
Andren et al.
Eur J Neurol. 2021
Probability of Survival
Time from baseline (years)
Number at risk

iNPH 改め "Hakim病"へ
idiopathic Normal Pressure Hydrocephalus
特発性
正常圧
水頭症
原因がはっきりしない
頭蓋内圧が正常のまま
頭に水が溜まる病気
脳室が拡大する機序
加齢（脳萎縮）
繊毛の脱落・機能低下
糖尿病、アルコール
頭蓋内圧の平均値は正常範囲だが、
脳のコンプライアンスは低下しており、
圧変動（振幅）は上昇している。



The NEW ENGLAND JOURNAL of MEDICINE
2025
September 16
ORIGINAL ARTICLE
A Randomized Trial of Shunting for Idiopathic Normal-Pressure Hydrocephalus
M.G. Luciano, M.A. Williams, M.G. Hamilton, H.L. Katzen, N.A. Dasher, A. Moghekar, J. Hua, J. Malm, A. Eklund, N. Alpert Abel, A.M. Raslan,
The Placebo-controlled Efficacy of iNPH Shunting (PENS) Trial



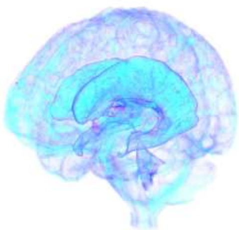
The NEW ENGLAND JOURNAL of MEDICINE
2025
December 3
REVIEW ARTICLE
Idiopathic Normal-Pressure Hydrocephalus
Mark D. Johnson, M.D., Ph.D., and Michael A. Williams, M.D.

idiopathic normal pressure hydrocephalus
Possible? Probable? iNPH
診断されない(見逃される）患者を減らしたい。
適切な治療を適切なタイミングに
多くの患者が享受できるようにしたい。

目標：Hakim病（iNPH）を適切に治療する

不確実な診断・治療となりやすい理由

1. 認知度が低い.
2. 手術しても治らないだろう.
3. 脳萎縮と見誤られる.

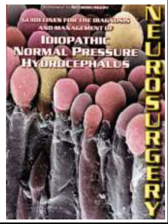


International Guidelines 2005

Anthony Marmarou, Norman Relkin, Marvin Bergsneider, Petra Klinge, **Peter Black**

Neurosurgery. 2005.

- **#1.** Development of guidelines for idiopathic normal-pressure hydrocephalus: Introduction
  Marmarou A, et al.
- **#2.** Diagnosing idiopathic normal-pressure hydrocephalus.
  Relkin N, et al.
- **#3.** The value of supplemental prognostic tests for the preoperative assessment of idiopathic normal-pressure hydrocephalus.
  Marmarou A, et al.
- **#4.** Surgical management of idiopathic normal-pressure hydrocephalus.
  Bergsneider M, et al.
- **#5.** Outcome of shunting in idiopathic normal-pressure hydrocephalus and the value of outcome assessment in shunted patients.
  Klinge P, et al.



2015

SPECIAL ARTICLE

AMERICAN ACADEMY OF NEUROLOGY.

Practice guideline: Idiopathic normal pressure hydrocephalus: Response to shunting and predictors of response

Report of the Guideline Development, Dissemination, and Implementation Subcommittee of the American Academy of Neurology

John J. Halperin, MD
Roger Kurlan, MD
Jason M. Schwalb, MD
Michael D. Cusimano, MD, PhD, FRCSC
Gary Gronseth, MD
David Gloss, MD

Correspondence to American Academy of Neurology: guidelines@aan.com

ABSTRACT

Objective: We evaluated evidence for utility of shunting in idiopathic normal pressure hydrocephalus (iNPH) and for predictors of shunting effectiveness.

Methods: We identified and classified relevant published studies according to 2004 and 2011 American Academy of Neurology methodology.

Results: Of 21 articles, we identified 3 Class I articles.

Conclusions: Shunting is possibly effective in iNPH (96% chance subjective improvement, 83% chance improvement on timed walk test at 6 months) (3 Class III). Serious adverse event risk was 11% (1 Class III). Predictors of success included elevated R_o (1 Class I, multiple Class II), impaired cerebral blood flow reactivity to acetazolamide (by SPECT) (1 Class I), and positive response to either external lumbar drainage (1 Class III) or repeated lumbar punctures. Age may not be a prognostic factor (1 Class II). Data are insufficient to judge efficacy of radionuclide cisternography or aqueductal flow measurement by MRI.

Recommendations: Clinicians may choose to offer shunting for subjective iNPH symptoms and gait (Level C). Because of significant adverse event risk, risks and benefits should be carefully weighed (Level B). Clinicians should inform patients with iNPH with elevated R_o and their families that they have an increased chance of responding to shunting compared with those without such elevation (Level B). Clinicians may counsel patients with iNPH and their families that (1) positive response to external lumbar drainage or to repeated lumbar punctures increases the chance of response to shunting, and (2) increasing age does not decrease the chance of shunting being successful (both Level C). Neurology® 2015;85:2063–2071

Neurology®

2016

Annals of NEUROLOGY
ANA AMERICAN NEUROLOGICAL ASSOCIATION®

EDITORIAL

裸の王様
The Emperor Has No Clothes

Clifford B. Saper,
Professor of Neurology,
Sleep Medicine,
Harvard Medical School

In the story, "The Emperor's New Clothes," a pompous Emperor is tricked by his tailor into accepting a suit made of a marvelous new fabric that is very sheer. The Emperor walks down the street proudly accepting compliments on his new clothes from his courtiers until a mere child sees him and cries out, "The Emperor has no clothes!"

I had a similar experience recently, when like most members of the American Academy of Neurology (AAN), I received an email from the AAN on December 7 proclaiming as a headline that "Ventricular Shunting Appears to Be Helpful in Patients with iNPH" (idio-

"upgraded the strength of evidence" for a positive effect "from very low to low" because 95% of subjects reported "subjective improvement" (although there was no control for a placebo response). The Guideline then goes on to assess whether various tests for cerebrospinal fluid (CSF) flow could predict which of the patients improved (a dubious proposition if there is "very weak" evidence for an overall effect, which may have been a placebo response).

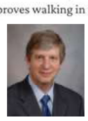
My own view is that the shunting procedure often works well if the patient has the classic triad of cognitive and gait impairment and incontinence due to hydroceph-

NeurologyToday®

Landmark Double-Blind Trial Confirms Shunt Benefit in iNPH

2026

"This is a landmark trial for the field. It's level-one evidence that surgery helps people with NPH," said Neil Graff-Radford, MD, endowed professor of neurology at the Mayo Clinic in Jacksonville, FL, who was not involved in the study. "It took an extraordinary number of years to pull off, and it finally shows in a controlled way that shunt surgery truly improves walking in these patients."



DR. MICHAEL A. WILLIAMS

"The only effective treatment for iNPH is sh… compare the risk/benefit profile of treating… surgery."

DR. CLIFFORD B. SAPER **Skepticism**

"We do not know if the improvement in gait persisted at six, 12, or 24 months. We need to continue rigorous study and long-term monitoring to understand which patients truly sustain improvement."

"I am afraid that the small clinical gain in gait speed demonstrated in this study will be used as a rationale for subjecting thousands of elderly people with dementia to shunting hoping that this will forestall the dementia, and instead leading to a great many useless surgeries and much needless morbidity, mortality, and expense."

Annals of NEUROLOGY
ANA AMERICAN NEUROLOGICAL ASSOCIATION®

2026

VIEWPOINT

Time to Reconsider the Value of Shunting Procedures for "Idiopathic Normal Pressure Hydrocephalus"

Clifford B. Saper, MD, PhD

Perhaps it is time for him to reconsider his own view…

The concept of "normal pressure hydrocephalus" dates back to the description by Hakim and Adams in 1965 of 2 series of patients with enlarged lateral ventricles, high normal cerebral spinal fluid (CSF) pressure and a triad of cognitive impairment, urinary incontinence, and gait apraxia. The validity of this concept is based upon the reversal of its symptoms by a shunting procedure. However, all of these patients had secondary communicating hydrocephalus after previous episodes of meningitis, brain trauma, or subarachnoid hemorrhage. Nevertheless, shunting procedures for "idiopathic normal pressure hydrocephalus" with similar symptoms but no antecedent cause for communicating hydrocephalus became widely performed, although they were not tested by randomized, placebo-controlled trials for their value until a series of recent publications. These trials do not find improvement in either cognitive function or continence after shunting, and show that there is a small (approximately 25%) improvement in gait speed. They also confirm earlier studies that these patients suffer an approximately 10 to 15% incidence of severe adverse events in the first year after shunting, which may further accumulate with time. In the absence of value for shunting in reversing cognitive impairment and incontinence, the entire concept of "idiopathic normal pressure hydrocephalus" is called into question.

ANN NEUROL 2026;00:1–6

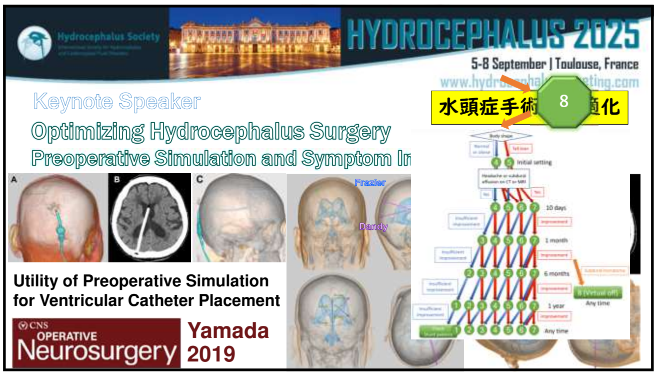
Subjective evaluation (improvement)

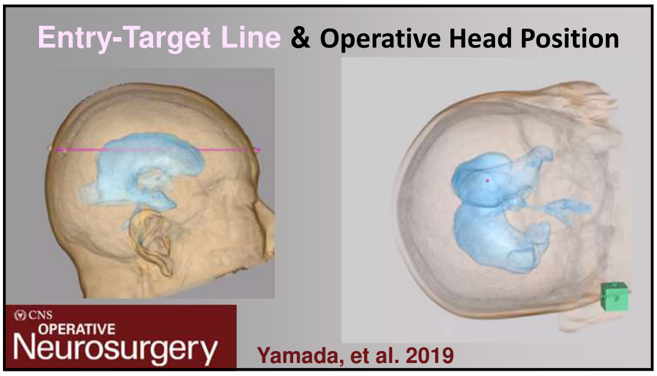
modified Rankin Scale
iNPH (various) Grading Scale
Gait features (short stepped, wide-based gait)
SCD-Q, CC-10, EQ-5D-5L, QABQ...

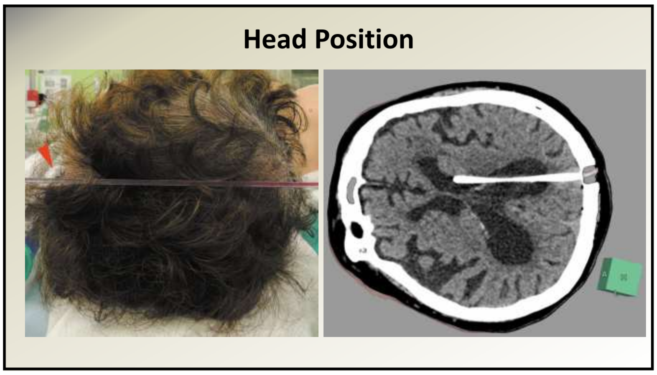
Quantitative measurement (improvement)

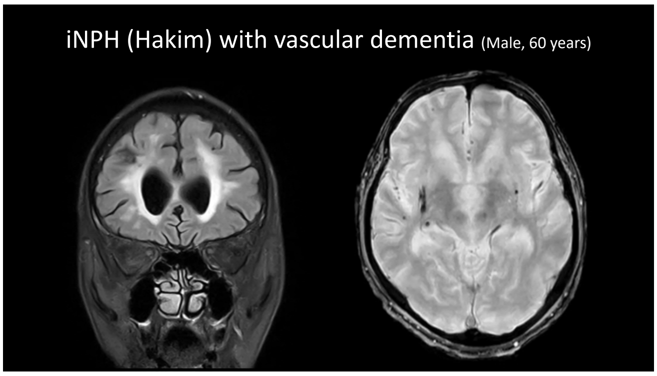
Timed Up & Go (TUG),
Gait speed, 10m-walk test
Tinetti (Gait & Balance), Berg Balance Scale
MMSE, MoCA, Stroop, RAVLT/AVLT

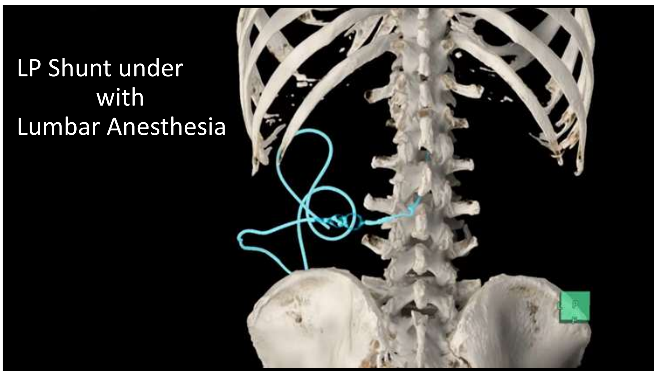
Comparable & Reliable!

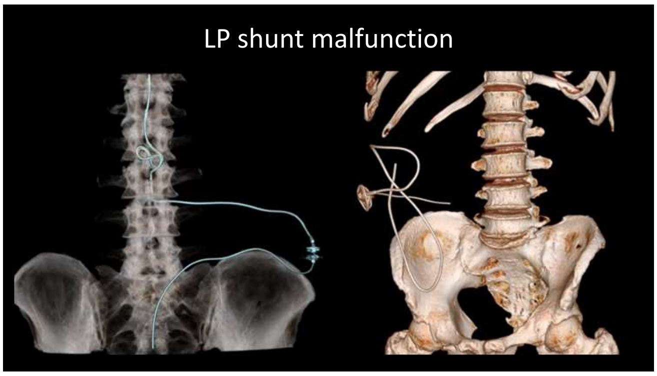


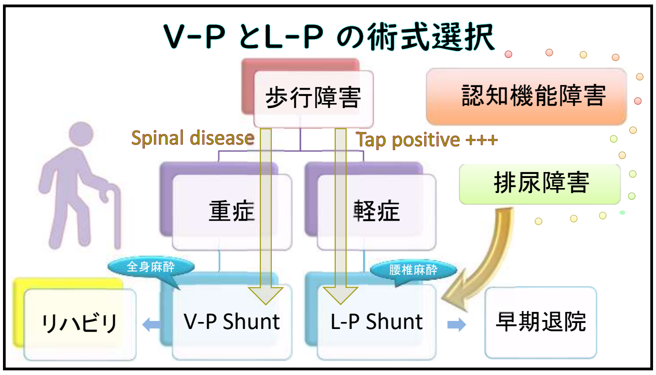












目標：Hakim病（iNPH）を適切に治療する

不確実な診断・治療となりやすい理由

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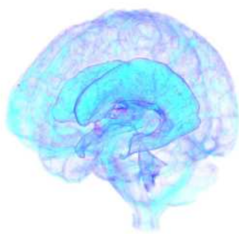
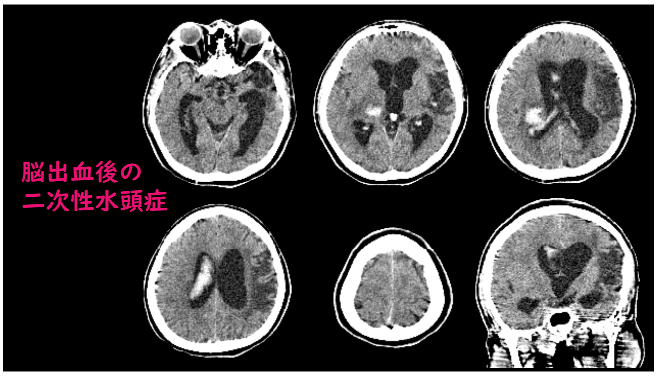
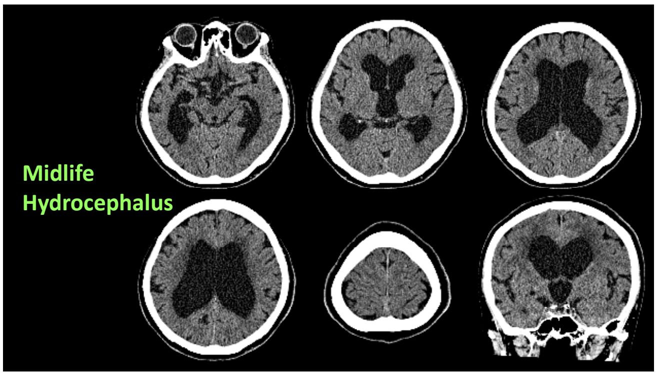
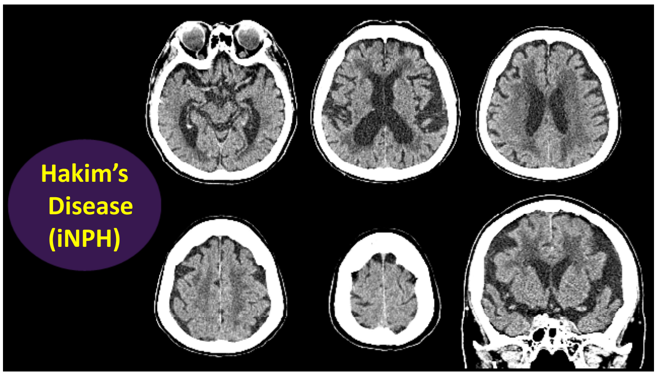
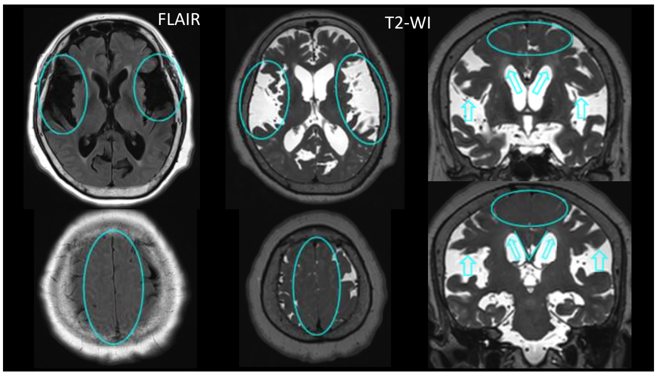
3. 脳萎縮と見誤られる。

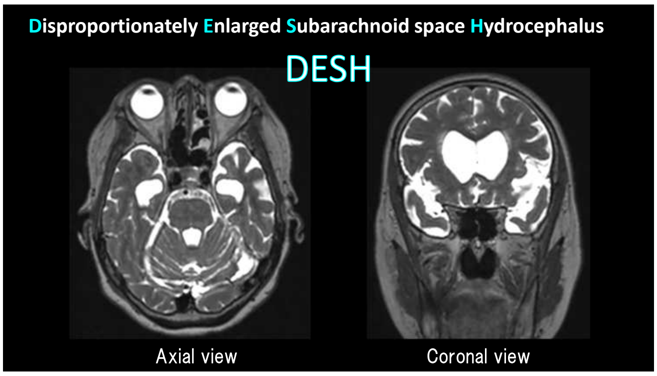


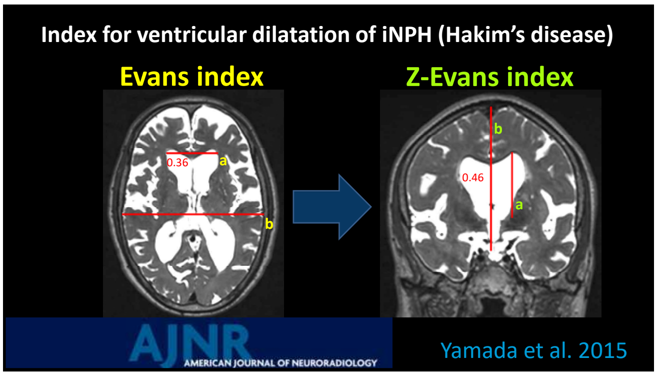


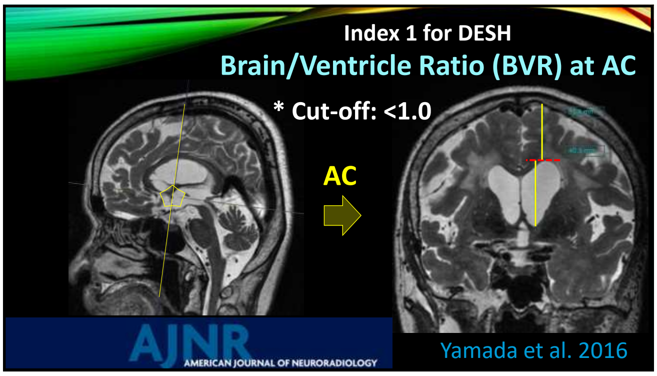


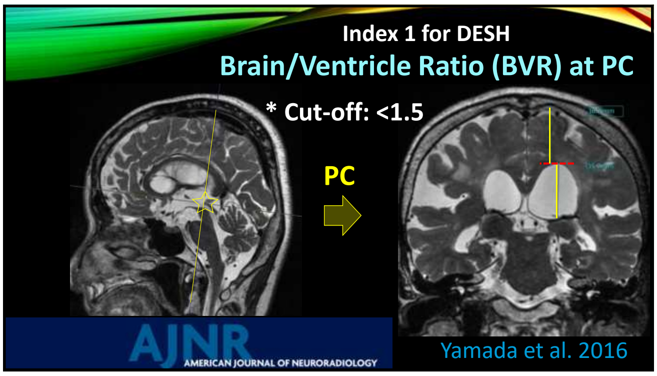

Hakim's
Disease
(iNPH)

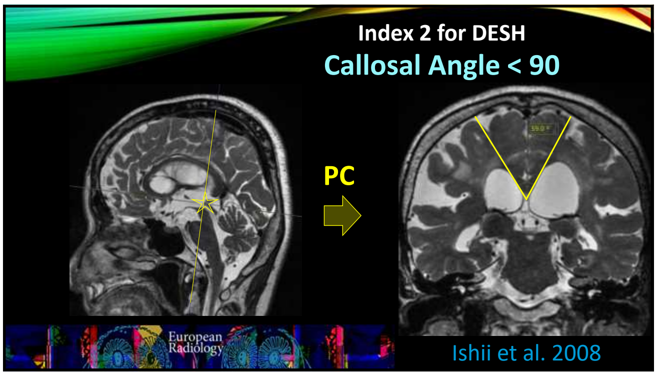

FLAIR
T2-WI

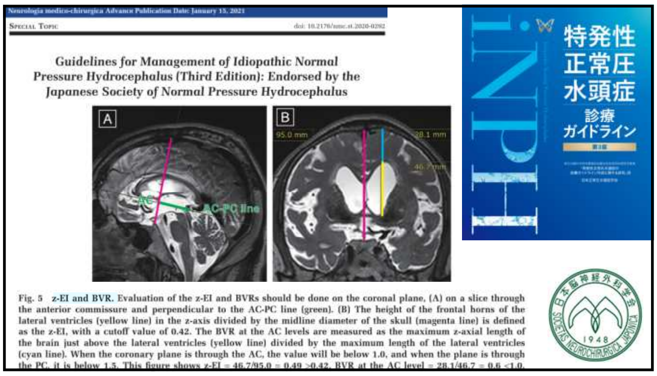

Disproportionately Enlarged Subarachnoid space Hydrocephalus
DESH
Axial view
Coronal view


Index for ventricular dilatation of iNPH (Hakim's disease)
Evans index
Z-Evans index
0.36
0.46
AJNR
AMERICAN JOURNAL OF NEURORADIOLOGY
Yamada et al. 2015


Index 1 for DESH
Brain/Ventricle Ratio (BVR) at AC
* Cut-off: <1.0
AC
AJNR
AMERICAN JOURNAL OF NEURORADIOLOGY
Yamada et al. 2016


Index 1 for DESH
Brain/Ventricle Ratio (BVR) at PC
* Cut-off: <1.5
PC
AJNR
AMERICAN JOURNAL OF NEURORADIOLOGY
Yamada et al. 2016

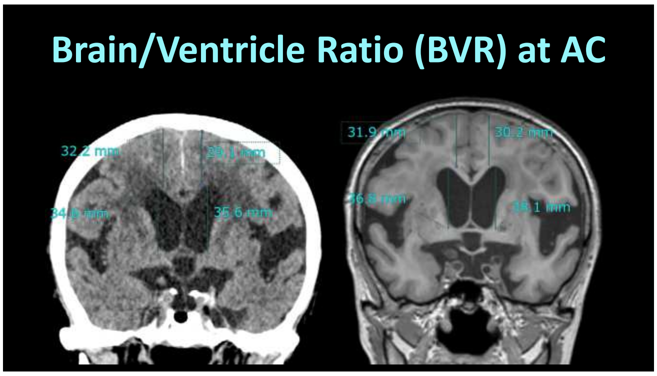


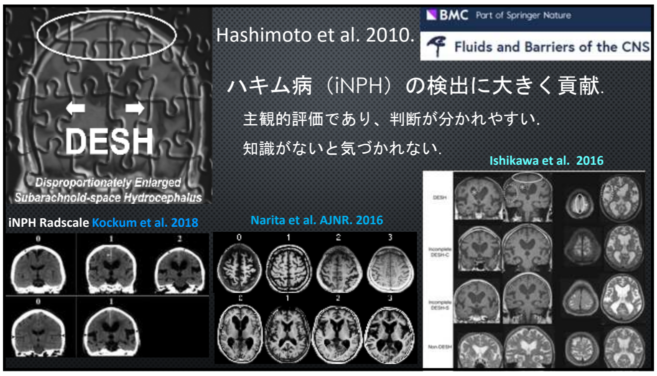

Neurologia medico-chirurgica Advance Publication Date: January 15, 2021

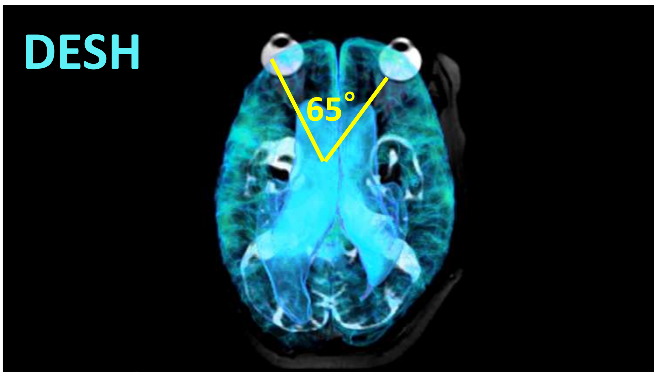
SPECIAL TOPIC

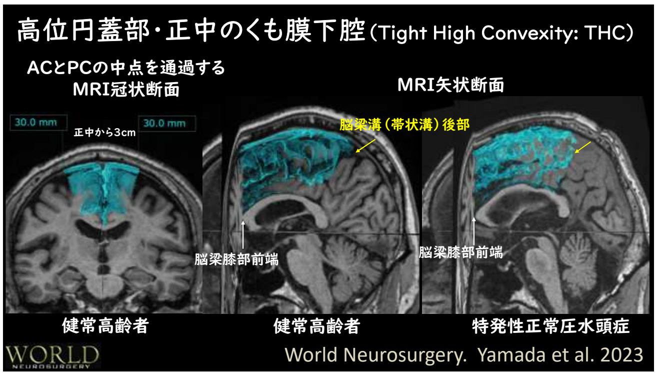
Guidelines for Management of Idiopathic Normal Pressure Hydrocephalus (Third Edition): Endorsed by the Japanese Society of Normal Pressure Hydrocephalus

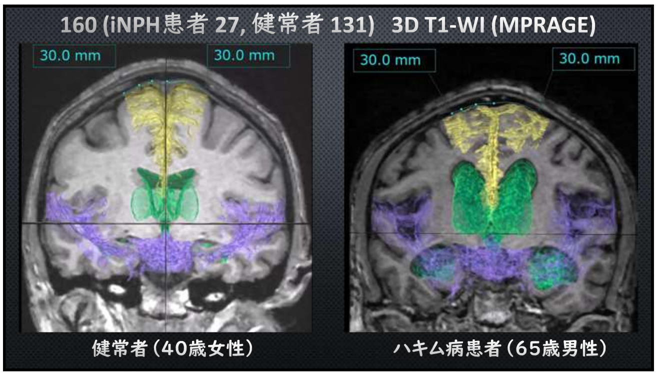
Fig. 5 z-EI and BVR. Evaluation of the z-EI and BVRs should be done on the coronal plane, (A) on a slice through the anterior commissure and perpendicular to the AC-PC line (green). (B) The height of the frontal horns of the lateral ventricles (yellow line) in the z-axis divided by the midline diameter of the skull (magenta line) is defined as the z-EI, with a cutoff value of 0.42. The BVR at the AC levels are measured as the maximum z-axial length of the brain just above the lateral ventricles (yellow line) divided by the maximum length of the lateral ventricles (cyan line). When the coronary plane is through the AC, the value will be below 1.0, and when the plane is through the PC, it is below 1.5. This figure shows z-EI = 46.7/95.0 = 0.49 >0.42. BVR at the AC level = 28.1/46.7 = 0.6 <1.0.

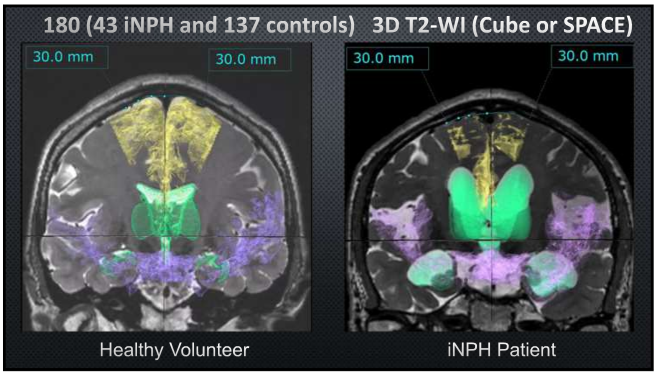


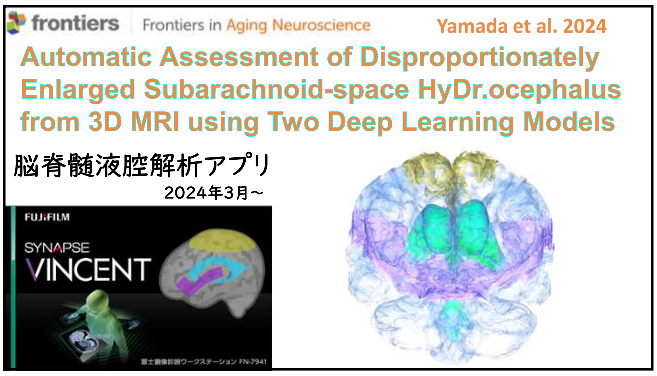


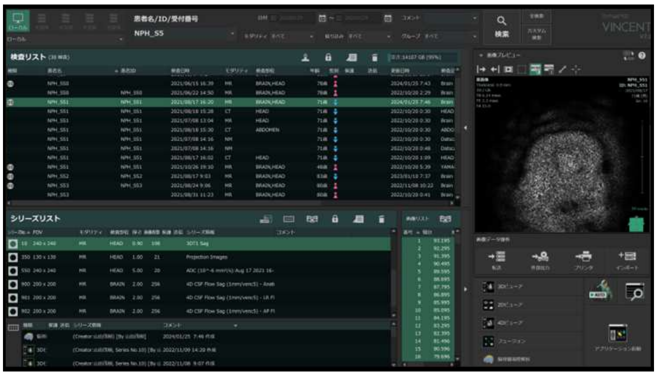


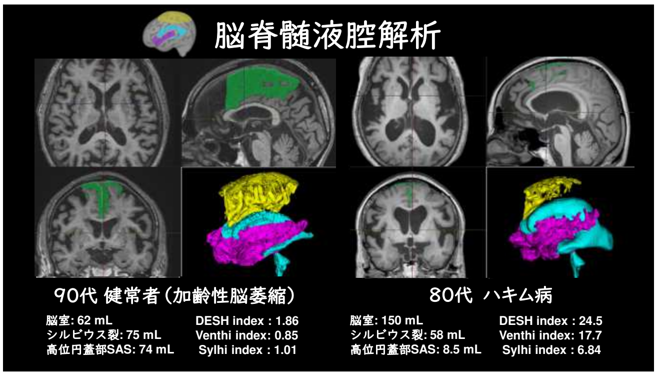

高位円蓋部・正中のくも膜下腔(Tight High Convexity: THC)
ACとPCの中点を通過する
MRI冠状断面
MRI矢状断面
30.0 mm
正中から3cm
30.0 mm
脳梁溝（帯状溝）後部
脳梁膝部前端
脳梁膝部前端
健常高齢者
健常高齢者
特発性正常圧水頭症
WORLD
World Neurosurgery. Yamada et al. 2023

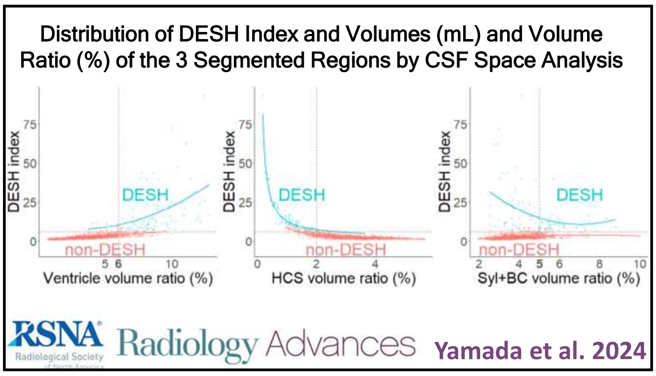

160 (iNPH患者 27, 健常者 131) 3D T1-WI (MPRAGE)
30.0 mm
30.0 mm
30.0 mm
30.0 mm
健常者（40歳女性）
ハキム病患者（65歳男性）

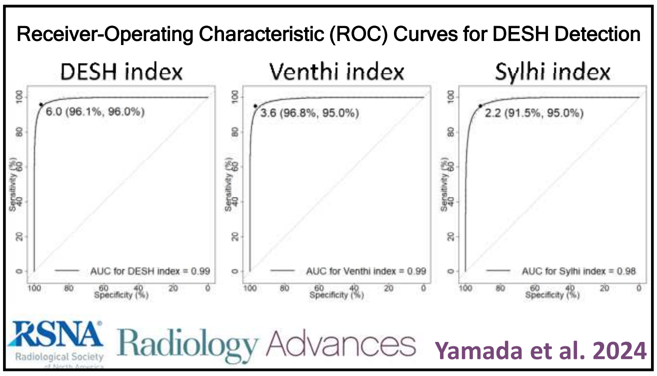

180 (43 iNPH and 137 controls) 3D T2-WI (Cube or SPACE)
30.0 mm
30.0 mm
30.0 mm
30.0 mm
Healthy Volunteer
iNPH Patient

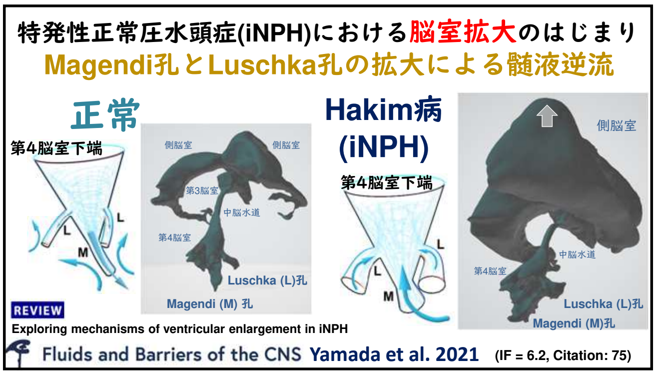

frontiers
Frontiers in Aging Neuroscience
Yamada et al. 2024
Automatic Assessment of Disproportionately Enlarged Subarachnoid-space HyDr.ocephalus from 3D MRI using Two Deep Learning Models
脳脊髄液腔解析アプリ
2024年3月～
FUJIFILM
SYNAPSE
VINCENT

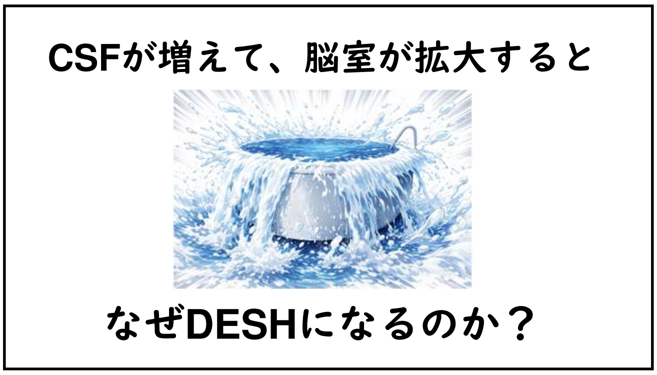


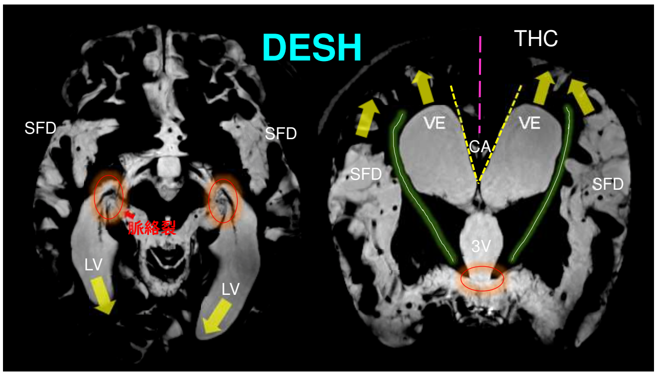

脳脊髄液腔解析
90代 健常者（加齢性脳萎縮）
脳室: 62 mL
シルビウス裂: 75 mL
高位円蓋部SAS: 74 mL
DESH index : 1.86
Venthi index: 0.85
Sylhi index : 1.01
80代 ハキム病
脳室: 150 mL
シルビウス裂: 58 mL
高位円蓋部SAS: 8.5 mL
DESH index : 24.5
Venthi index: 17.7
Sylhi index : 6.84

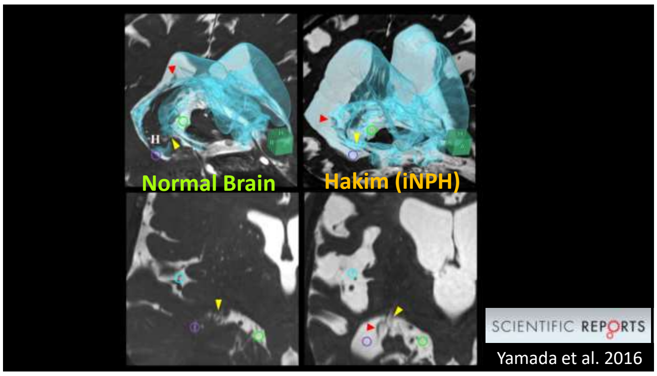

Distribution of DESH Index and Volumes (mL) and Volume Ratio (%) of the 3 Segmented Regions by CSF Space Analysis
DESH index
DESH
non-DESH
Ventricle volume ratio (%)
HCS volume ratio (%)
Syl+BC volume ratio (%)
RSNA
Radiology Advances
Yamada et al. 2024

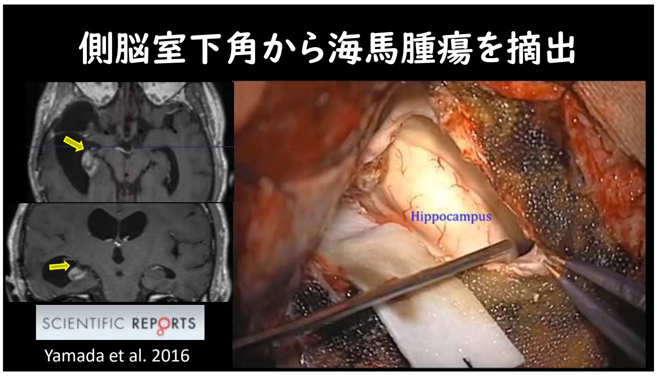

Receiver-Operating Characteristic (ROC) Curves for DESH Detection
DESH index
Venthi index
Sylhi index
6.0 (96.1%, 96.0%)
3.6 (96.8%, 95.0%)
2.2 (91.5%, 95.0%)
AUC for DESH index = 0.99
AUC for Venthi index = 0.99
AUC for Sylhi index = 0.98
Specificity (%)
RSNA
Radiology Advances
Yamada et al. 2024

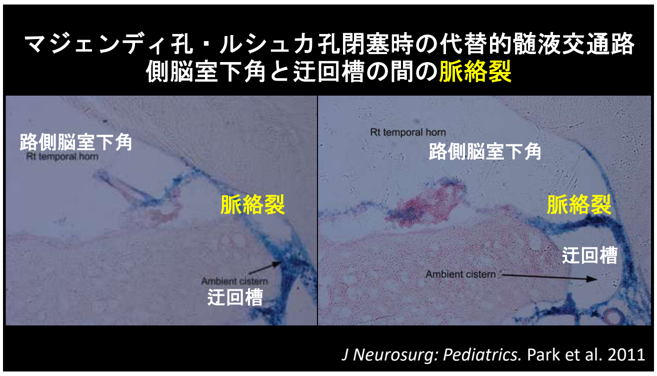

特発性正常圧水頭症(iNPH)における脳室拡大のはじまり
Magendi孔とLuschka孔の拡大による髄液逆流
正常
第4脳室下端
側脳室
第3脳室
中脳水道
第4脳室
Luschka (L)孔
Magendi (M) 孔
Hakim病
(iNPH)
REVIEW
Exploring mechanisms of ventricular enlargement in iNPH
Fluids and Barriers of the CNS Yamada et al. 2021
(IF = 6.2, Citation: 75)

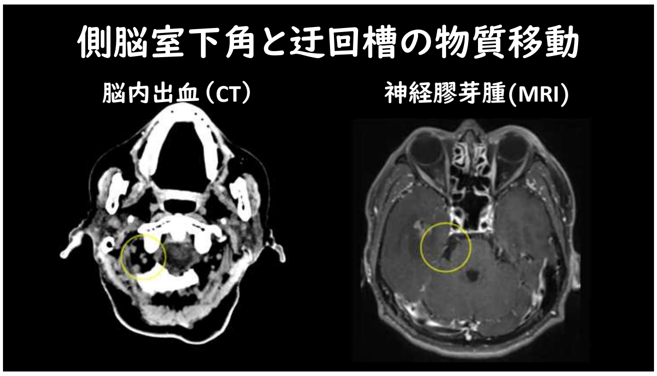

CSFが増えて、脳室が拡大すると
なぜDESHになるのか？

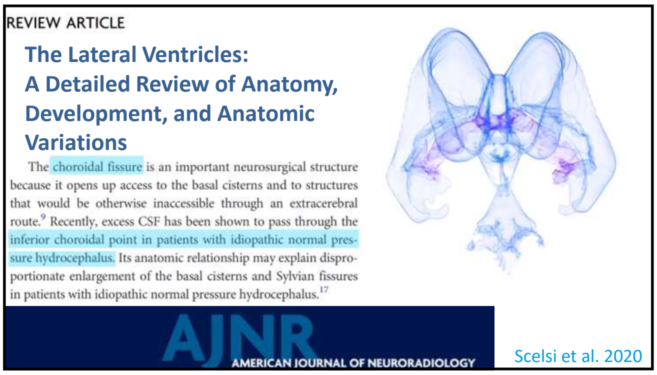

DESH
THC
SFD
LV
脈絡叢
VE
CA
3V

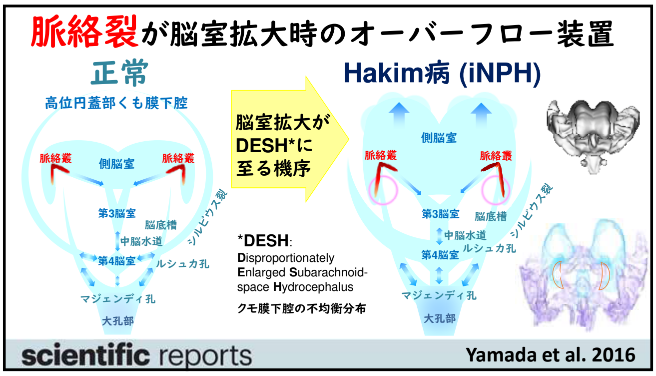

Normal Brain
Hakim (iNPH)
SCIENTIFIC REPORTS
Yamada et al. 2016

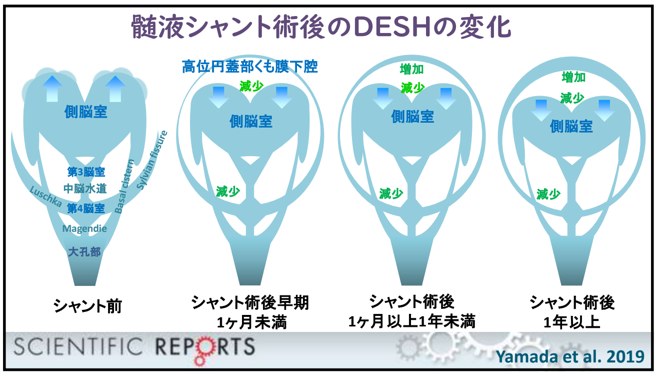

側脳室下角から海馬腫瘍を摘出
Hippocampus
SCIENTIFIC REPORTS
Yamada et al. 2016


マジェンディ孔・ルシュカ孔閉塞時の代替的髄液交通路
側脳室下角と迂回槽の間の脈絡裂
路側脳室下角
Rt temporal horn
脈絡裂
Ambient cistern
迂回槽
Rt temporal horn
路側脳室下角
脈絡裂
迂回槽
Ambient cistern
J Neurosurg: Pediatrics. Park et al. 2011


側脳室下角と迂回槽の物質移動
脳内出血（CT）
神経膠芽腫(MRI)


REVIEW ARTICLE
The Lateral Ventricles:
A Detailed Review of Anatomy, Development, and Anatomic Variations
The choroidal fissure is an important neurosurgical structure because it opens up access to the basal cisterns and to structures that would be otherwise inaccessible through an extracerebral route.[9] Recently, excess CSF has been shown to pass through the inferior choroidal point in patients with idiopathic normal pressure hydrocephalus. Its anatomic relationship may explain disproportionate enlargement of the basal cisterns and Sylvian fissures in patients with idiopathic normal pressure hydrocephalus.[17]
AJNR AMERICAN JOURNAL OF NEURORADIOLOGY
Scelsi et al. 2020


脈絡裂が脳室拡大時のオーバーフロー装置
正常
高位円蓋部くも膜下腔
脈絡叢
側脳室
脈絡叢
第3脳室
脳底槽
中脳水道
シルビウス裂
第4脳室
ルシュカ孔
マジェンディ孔
大孔部
脳室拡大がDESH*に至る機序
*DESH:
Disproportionately Enlarged Subarachnoid-space Hydrocephalus
クモ膜下腔の不均衡分布
Hakim病 (iNPH)
側脳室
脈絡叢
脈絡叢
第3脳室
脳底槽
中脳水道
シルビウス裂
第4脳室
ルシュカ孔
マジェンディ孔
大孔部
scientific reports
Yamada et al. 2016


髄液シャント術後のDESHの変化
高位円蓋部くも膜下腔
減少
増加
減少
増加
減少
側脳室
側脳室
側脳室
側脳室
第3脳室
中脳水道
第4脳室
Luschka
Magendie
大孔部
Basal cistern
Sylvian fissure
減少
減少
減少
シャント前
シャント術後早期
1ヶ月未満
シャント術後
1ヶ月以上1年未満
シャント術後
1年以上
SCIENTIFIC REPORTS
Yamada et al. 2019